DATA ANALYTICS PROJECT

Adventure Work

Sales Performance Analysis

Excel • Tableau • Power BI • MySQL

GROUP - 01

PROJECT TEAM

Group 01 – Adventure Works

1. M. Posibabu
2. Savita Gowtam
3. R. Jagadeesh
4. S. Bhaskar Sai Charan

PROJECT OVERVIEW

About Adventure Works Cycles

Adventure Works Cycles is a multinational bicycle manufacturer operating across North American, European and Asian markets, headquartered in Bothell, Washington.

Data Modeling

Star schema with Fact_Sales linked to dimension tables

MySQL Queries

Questions answered using joins, date functions and KPIs

Visualizations

Excel, Tableau and Power BI Dashboards

Model : Fact_Sales -> DimCustomer | DimProduct | DimDate | DimSalesTerritory

29.36M	12.08M	17M	60K	41.15%
Total Sales	Profit	Production cost	Total Orders	Profit Margin

EXCEL DASHBOARD

Microsoft Excel | pivot Tables, charts & Slicers

Microsoft Excel

ADVENTURE WORK DASHBOARD

Sales
$29,358.7K

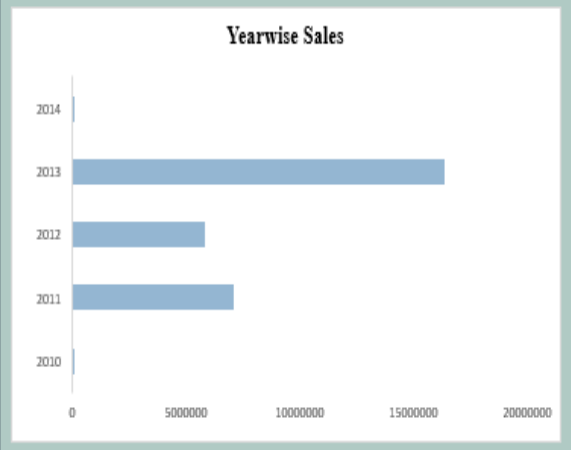
Profit
$12,080.9K

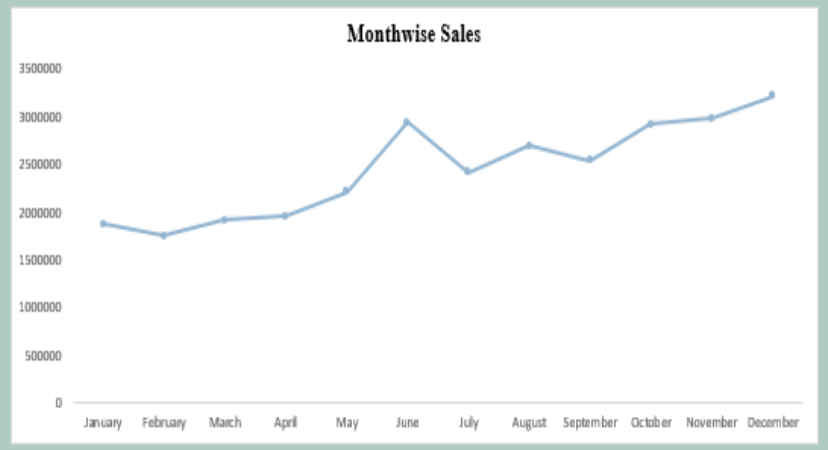
Order Quantity
60,398

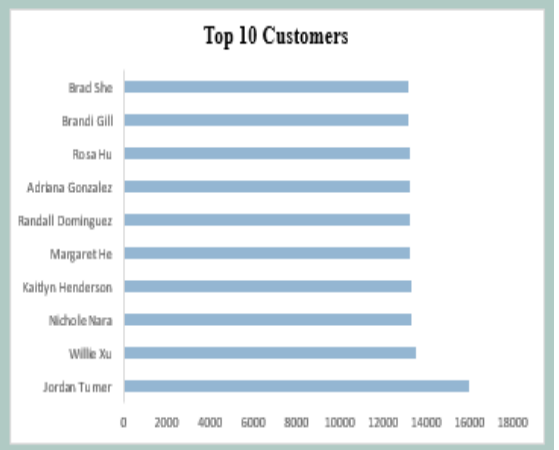
Mont...
January
February
March
April

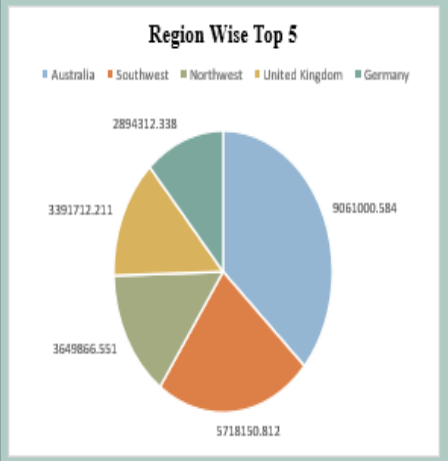
Year
2010
2011
2012
2013
2014

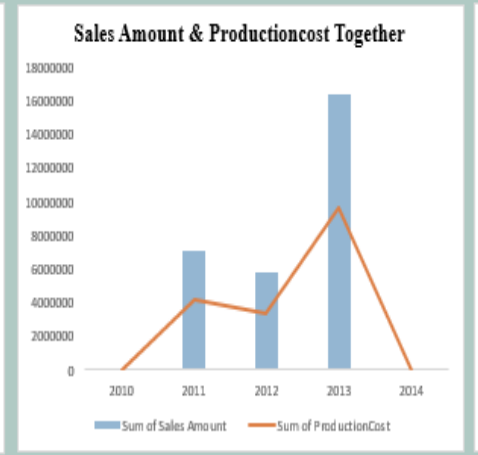
Produ...
Long-Sle...
Long-Sle...
Long-Sle...
ML Mount...

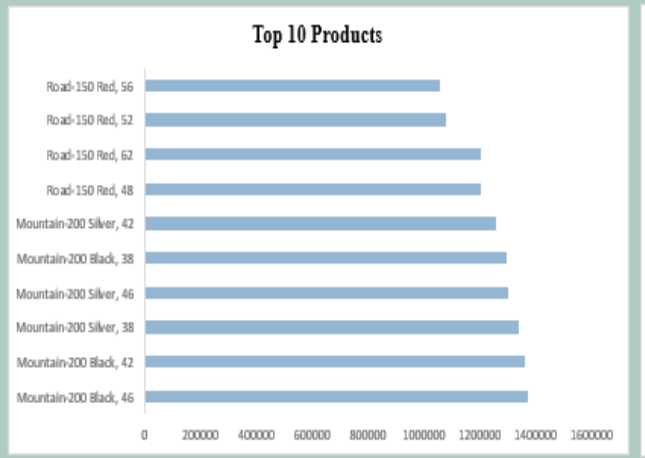


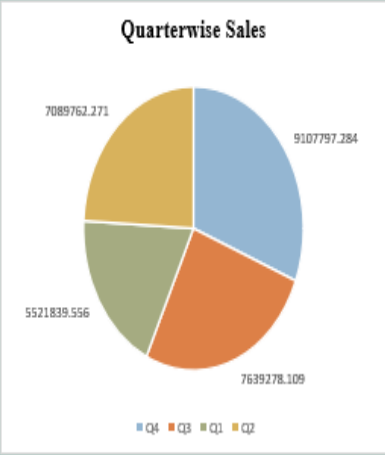












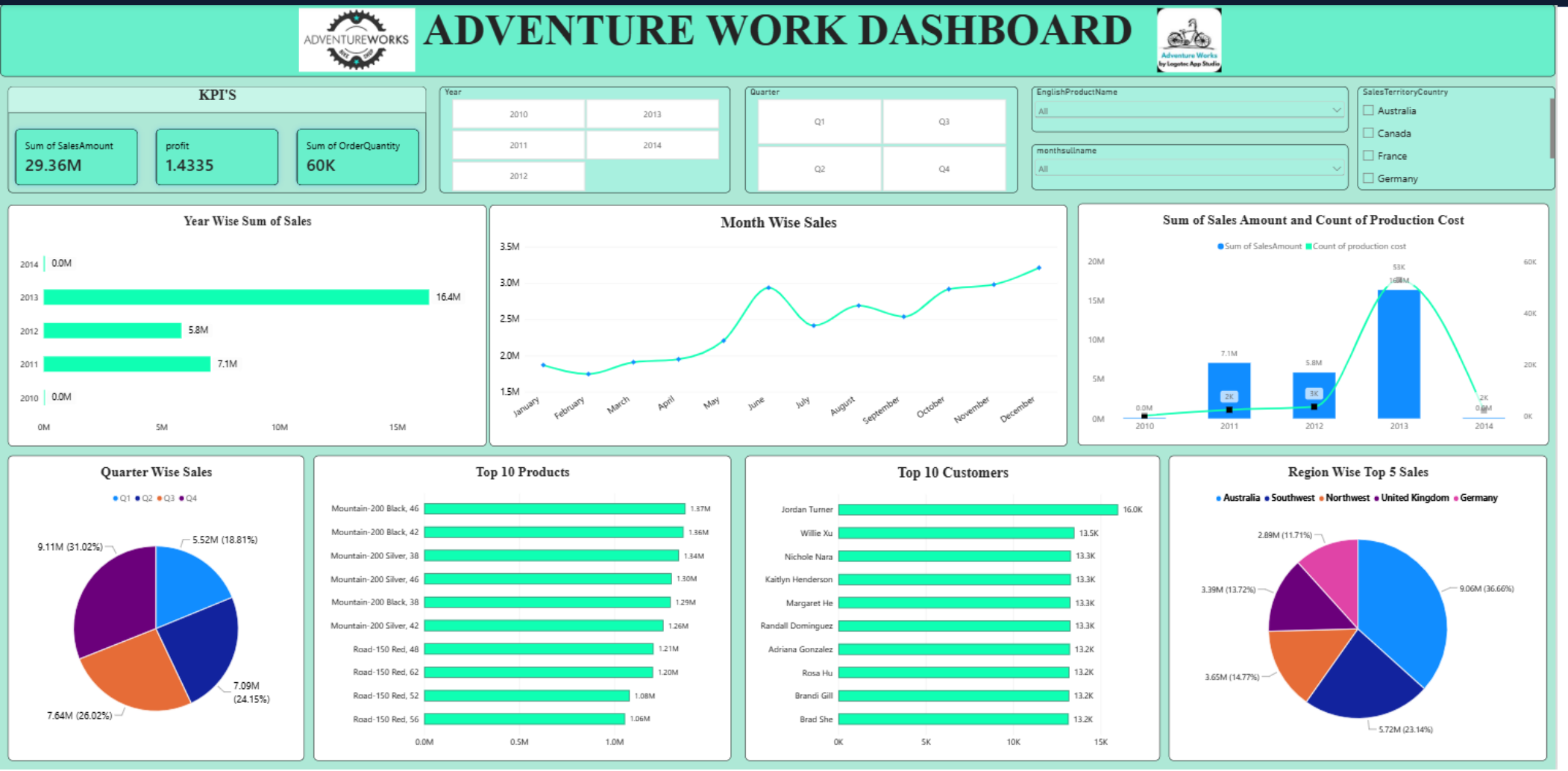
- Year Wise Sales
- Quarter Pie chart
- Region Sales Map
- Top 10 Customers
- Top 10 Products
- Sales Vs Prod cost

✅ Includes Year, Region & Category Slicers for interactive Filtering

POWER BI DASHBOARD – CUSTOMER ANALYSIS

Microsoft Power BI | Customer Revenue & Segmentation

Microsoft Power BI



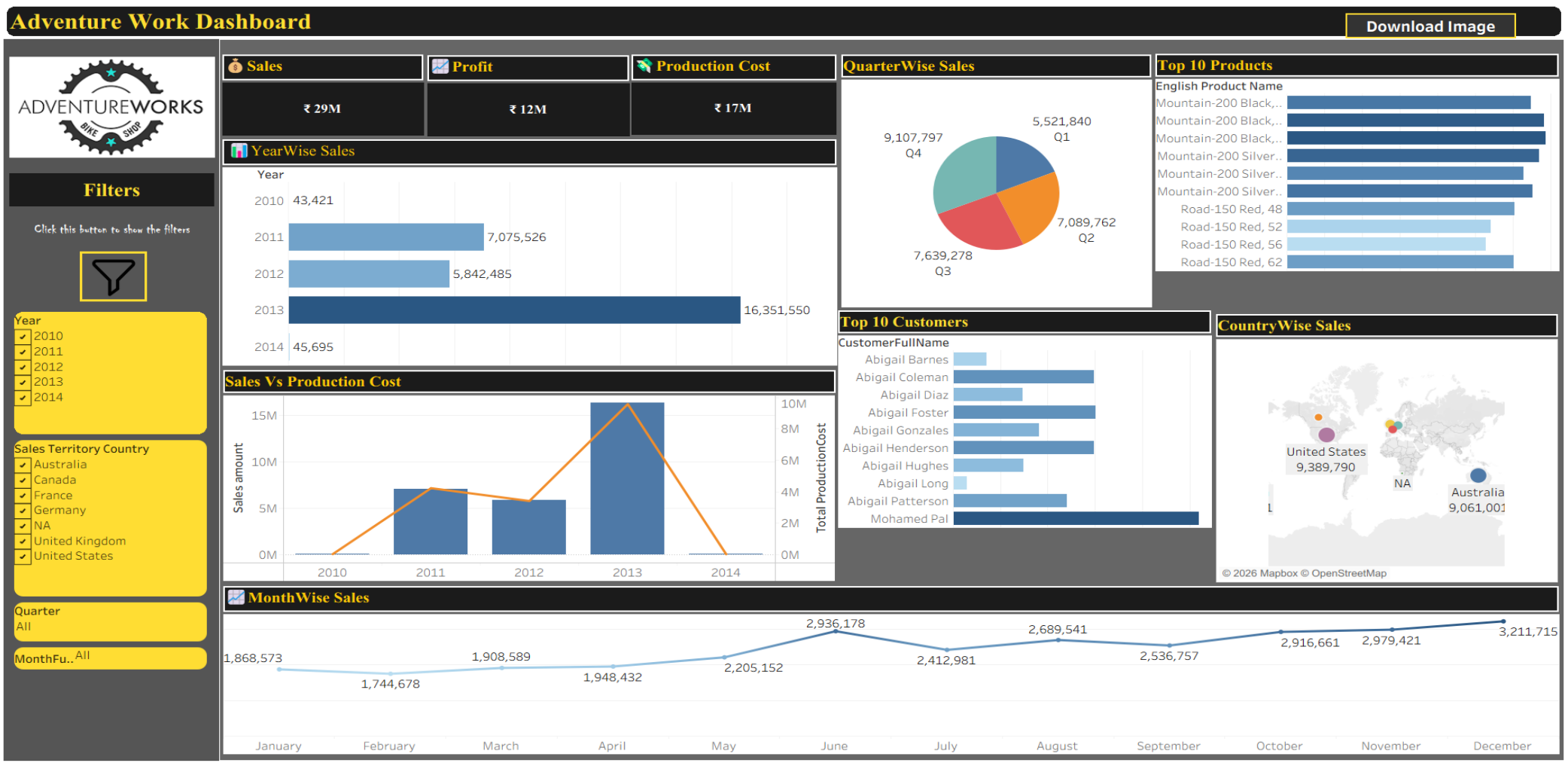
- Year Wise Sales
- Quarter Pie chart
- Region Sales Map
- Top 10 Customers
- Top 10 Products
- Sales Vs Prod cost

✅ Includes Year, Month, Subcategory, Country & Customer filters with drill down

TABLEAU DASHBOARD

Tableau Desktop | Sales Analysis & Customer Analysis

Tableau Dashboard



- Year Wise Sales
- Quarter Pie chart
- Region Sales Map
- Top 10 Customers
- Top 10 Products
- Sales Vs Prod cost

✅ Includes Year, Month, Subcategory, Country & Customer filters with drill down

MySQL QUERIES -- Q0 To Q6

MySQL | Data Transformation & Calculated Columns

Q0 . Union of both fact tables

```
CREATE or replace  view combined_fact_tables AS
 SELECT * FROM factinternetsales
 UNION ALL
 SELECT * FROM fact_internet_sales_new;
```

Q1 . Product Name and Unit price From Product Table

```
select s.*,
p.EnglishProductName as ProductName,
p.`Unit price` as Unitprice from combined_fact_tables s left join dimproduct p
on s.productkey = p.productkey;
```

Q2 . Customer full Name From Customer table

```
select s.*,
concat(c.firstname, ' ' ,c.lastname) as CustomerFullName from combined_fact_tables s
left join dimcustomer c
on s.customerkey = c.customerkey;
```

Q3 . Date Column From OrderDateKey

```
YEAR(...) AS Year, MONTH(...) AS MonthNo, MONTHNAME(...) AS MonthName,
CONCAT('Q', QUARTER(...)) AS Quarter, DATE_FORMAT(...,'%Y-%b') AS YearMonth,
DAYOFWEEK(...) AS WeekdayNo, DAYNAME(...) AS WeekdayName,
CASE WHEN MONTH(...) >= 4 THEN MONTH(...) - 3 ELSE MONTH(...) + 9 END AS
FinancialMonth,
CASE WHEN MONTH(...) BETWEEN 4 AND 6 THEN 'Q1' ... END AS FinancialQuarter
```

Q4 . Sales Amount

```
SELECT
(UnitPrice * OrderQuantity * (1 - UnitPriceDiscountPct)) AS SalesAmount
FROM combined_fact_tables
```

Q5 . Production Cost

```
SELECT (ProductStandardCost * OrderQuantity) AS ProductionCost
FROM combined_fact_tables;
```

Q6 . Profit

```
SELECT (Sales Amount )  -  ( production cost ) AS Profit
FROM combined_fact_tables;
```

MySQL QUERIES – Q7 To Q13

MySQL | Aggregations, Charts Data & KPI Summary

Q7 . Monthly Sales (pivot Table)

```
select
monthname(str_to_date(cast(OrderDatekey as char), '%Y%m%d')) as
Monthname,sum(SalesAmount) as Total_salesfrom combined_fact_tables
group by monthname , month(str_to_date(cast(OrderDatekey as char), '%Y%m%d'))
 order by month (str_to_date(cast(OrderDatekey as char), '%Y%m%d')) ;
```

Q8 . Year wise Sales (Bar chart)

```
Select
year (str_to_date(cast(OrderDateKey as char), '%Y%m%d')) as year,
sum(SalesAmount) as Total_Sales from combined_fact_tables
group by year order by year;
```

Q9 . Month Wise Sales (line chart)

```
select
monthname(str_to_date(cast(OrderDatekey as char), '%Y%m%d')) as
Monthname,sum(SalesAmount) as Total_salesfrom combined_fact_tables
group by monthname , month(str_to_date(cast(OrderDatekey as char), '%Y%m%d'))
 order by month (str_to_date(cast(OrderDatekey as char), '%Y%m%d')) ;
```

Q10 . Quarter wise Sales (pie chart)

```
Select
 quarter(str_to_date(cast(orderdatekey as char), '%Y%m%d')) as quarter, sum(SalesAmount)
as Total_sales from combined_fact_tables
 group by quarterorder by quarter;
```

Q11 . Combination Chart (sales & production cost)

```
WITH base AS (…) AS OrderDate, SalesAmount, TotalProductCost
  FROM combined_fact_tables)
SELECT    YEAR(OrderDate) AS Year, ROUND(SUM(SalesAmount), 2) AS Total_sales,
ROUND(SUM(TotalProductCost), 2) AS Production_costFROM baseGROUP BY
YEAR(OrderDate)ORDER BY YEAR(OrderDate);
```

Q12 (1) . Top 10 Products

```
selectp.englishproductname AS productname,sum(s.SalesAmount) as Total_sales from
combined_fact_tables s left join dimproduct p on s.productkey = p.ProductKey group by
p.EnglishProductName order by Total_sales desc limit 10;
```

Q12 (2). Top 10 customers

```
select concat(c.firstname,' ',c.lastname) as CustomerFullName,sum(s.SalesAmount) as
Total_salesfrom combined_fact_tables sleft join dimcustomer c on s.customerkey =
c.customerkey group by c.customerkey,c.FirstName,c.LastNameorder by Total_sales desc
limit 10;
```

Q12 (3) . Region Wise Sales

```
select t.salesterritoryregion,sum(s.SalesAmount) as Total_sales from combined_fact_tables s
left join dimsalesterritory t on s.salesterritorykey = t.SalesTerritoryAlternateKey group by
t.SalesTerritoryKey , t.SalesTerritoryRegionorder by Total_sales desclimit 5;
```

MySQL QUERIES – Q13

MySQL | Aggregations, Charts Data & KPI Summary

Q13 . Overall Dashboard KPIs

```
SELECT
  SUM(s.orderquantity * s.unitprice) AS Total_Sales_Amount,
   SUM(s.orderquantity * s.totalproductcost) AS Total_Production_Cost,
 SUM((s.orderquantity * s.unitprice) - (s.orderquantity * s.totalproductcost)) AS Total_ProfitFROM
combined_fact_tables s;
```

KEY INSIGHTS

Adventure work Sales Analysis | 2010 - 2014

29.36M	12.08M	17M	60K	41.15%
Total Sales	Total Profit	Production Cost	Total Orders	Profit margin

1 Peak Year : 2013

2013 was the highest sales year at 16M, 2.7x higher than 2012. Significant growth from 2011 onward.

2 Top Region : Southwest

Southwest USA led all regions. Australia was the top international market with 9.06M in sales

3 Top Products

Mountain-200 Black (42) was the best seller at 1.37M. Mountain bikes dominate the top 10 products.

4 Top Customers

Jordan Turner leads with 10,000 in purchases. Top 10 customers are very closely ranked between 13-16K.

5 Peaks Sales

Q4 is consistently the strongest quarter, with a holiday season effect visible across all years.

6 Sales > Cost

Production cost stays well below sales amount across all months, giving a healthy and stable margin of 41.15%.

THANK YOU

Adventure Works – Sales Performance Analysis

Excel • Tableau • Power BI • MySQL

GROUP - 01

Posibabu | Savita Gowtham | Jagadeesh | Bhaskar Sai Charan